Superelliptic generalisation of the theory of hidden vanishings:

the family $y^n = x^d + \sigma x^r + 1$

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Abstract

We extend the theory of hidden vanishings of the Cartier–Manin matrix, established for the hyperelliptic family $y^2 = x^d + \sigma x^r + 1$, to the superelliptic families $y^n = x^d + \sigma x^r + 1$ ($n \geq 2$, $\gcd(d, n) = 1$) over $\mathbb{F}_p$ with $\gcd(p, n) = 1$. Two results are established. **Theorem A** constructs, with a complete proof from the axioms of the Cartier operator, the generalised Cartier matrix with respect to the basis of differentials $\omega_{i,k} = x^{i-1}\,dx/y^k$ ($1 \leq i \leq g_k$): its blocks, indexed by $k = 1, \ldots, n-1$, are linked by $k \mapsto k_{\text{out}} \equiv k\,p^{-1} \pmod n$, with multinomial exponent $N(k) = (k_{\text{out}}\,p - k)/n$ and entries given by coefficient extraction in $f(x)^{N(k)}$. **Theorem B** establishes that the p-rank of the Jacobian is *not* the rank of this matrix after a single application, but its **stabilised rank under iteration** (Fitting's lemma) — a point we first missed, which produced reproducible numerical discrepancies, and which we correct here with a proof and a complete example. Everything is checked independently by point counting on 18 configurations covering $n = 2, 3, 4$ and both regimes $p \equiv 1 \pmod n$ and $p \not\equiv 1 \pmod n$. A further part (§7–8) proves by hand a description of the kernel of the blocks, a duality law and a criterion for $d = p$, and classifies 143 cases of hidden vanishings (status [D]/[V]/[O]); 44 cases remain unexplained. Sections §9–10 add a general Menger theorem ($\Delta < g_0 \Rightarrow \det \Phi \equiv 0$), its closed criterion proved for $d = p$ (checked on 6104 cases) and the decomposition of the observed mechanisms. The tallies quoted come from distinct sweeps (different ranges of p, d) and are not reconciled with one another. Version 4 adds (§11–15) a proved degree law, two examples that can be checked entirely by hand (one cancellation and one Fermat-type case, $p = 3$), a multiplicity/corank theorem, a survey of the corank outside the family $p \equiv -1 \pmod n$, Conjecture G (generic support) and a *coverage correction*: the symmetry $r \leftrightarrow d - r$, which some recent sweeps assumed, is false. Version 4.1 (§16) adds: the second-iterate lemma (for $\sigma \in \mathbb{F}_p$), the description of the kernel of a block as the image of a first-order differential operator (**[D]**), a recurrence between columns explaining the twin columns, numerical laws for the profile of the exact forms (**[V]**), a rank bound by cuts, the census of 27 distinct cases of exact cancellation (instead of 33 with duplicates) and a complete by-hand proof of one mixed case. Version 4.2 adds a test of the variant $y^n = x^d + \sigma x^r + t$ and of values of the parameter outside $\mathbb{F}_p$ (§17).

1 Setting and notation

Let p be an odd prime, $n \geq 2$ with $\gcd(p, n) = 1$, and

$$\mathcal{C} \;:\; y^n = f(x) = x^d + \sigma x^r + 1, \qquad 0 < r < d,\ \gcd(d, n) = 1,$$

a smooth curve over $\mathbb{F}_p$ (we assume f has no multiple root). The condition $\gcd(d, n) = 1$ ensures total ramification (index n) at the unique point at infinity P_∞, which is therefore automatically $\mathbb{F}_p$-rational: being the only geometric point above $x = \infty$, it is necessarily fixed by Frobenius.

By Riemann–Hurwitz (total ramification at the d roots of f and at P_∞, i.e. $d+1$ branch points), the genus is

$$g = \frac{(n-1)(d-1)}{2}.$$

The space $H^0(\mathcal{C}, \Omega^1)$ has as a basis the differentials (a classical fact for cyclic covers of $\mathbb{P}^1$; see e.g. Hase-Liu and Triantafillou, *Efficient Point-Counting Algorithms for Superelliptic Curves*, arXiv:1709.02442)

$$\omega_{i,k} = \frac{x^{i-1}}{y^k}\,dx, \qquad 1 \le k \le n-1, \quad 1 \le i \le g_k, \qquad g_k := \left\lfloor \frac{kd}{n} \right\rfloor,$$

and $\sum_{k=1}^{n-1} g_k = g$.

2 Theorem A: the generalised Cartier operator

We recall the three properties that characterise the Cartier operator $C : H^0(\Omega^1) \to H^0(\Omega^1)$ on a smooth curve over $\mathbb{F}_p$ (Cartier, 1957; see Serre, *Sur la topologie des variétés algébriques en caractéristique p*): for every function h of the function field and every integer s,

(i) $C(h^p\,\omega) = h\,C(\omega)$; (ii) $C(x^s\,dx) = 0$ if $s \not\equiv p-1 \pmod p$; (iii) $C(x^{p-1}\,dx) = dx$.

These three axioms determine C uniquely on every subspace spanned by differentials of the form $g(x)\,dx$ with $g \in \mathbb{F}_p(x)$: writing $g(x) = \sum_m c_m x^m$, additivity and (i) applied to $h = x^{j-1}$, with $x^m\,dx = (x^{j-1})^p \cdot x^{p-1}dx$ for $m = pj-1$, give the classical formula

$$C\Big(\sum_m c_m x^m\,dx\Big) = \sum_{j\ge 1} c_{pj-1}^{1/p}\,x^{j-1}\,dx. \tag{$\star$}$$

Lemma 2.1 (Inversion of y^k)**.** *For $1 \le k \le n-1$, let $k_{\mathrm{out}} \in \{1,\dots,n-1\}$ be the unique integer with $k_{\mathrm{out}}\,p \equiv k \pmod n$, and set*

$$N(k) = \frac{k_{\mathrm{out}}\,p - k}{n} \;\in\; \mathbb{Z}_{\ge 0}.$$

Then, in the function field of $\mathcal{C}$,

$$\frac{1}{y^k} = f(x)^{N(k)} \cdot \Big(\frac{1}{y^{k_{\mathrm{out}}}}\Big)^p.$$

Proof. By construction $k_{\mathrm{out}}\,p - k = nN(k)$, so $y^{nN(k)} = y^{k_{\mathrm{out}}p-k}$, i.e. $y^{nN(k)} \cdot y^k = y^{k_{\mathrm{out}}p}$. Since $y^n = f(x)$, the left-hand side equals $f(x)^{N(k)} \cdot y^k$, hence $f(x)^{N(k)} \cdot y^k = y^{k_{\mathrm{out}}p} = (y^{k_{\mathrm{out}}})^p$. Dividing both sides by $y^k \cdot (y^{k_{\mathrm{out}}})^p$ gives the claimed equality. (k_{out} exists and is unique because p is invertible modulo n by the hypothesis $\gcd(p,n) = 1$; $N(k) \ge 0$ because $k_{\mathrm{out}} \ge 1$, so $k_{\mathrm{out}}p \ge p \ge k$ as soon as $p \ge n$, a condition satisfied in all our uses; $N(k) < p$ because $k_{\mathrm{out}} < n$.) □

Theorem 2.2 (Generalised Cartier operator)**.** *For $1 \le k \le n-1$ and $1 \le i \le g_k$,*

$$C(\omega_{i,k}) = \sum_{i'=1}^{g_{k_{\mathrm{out}}}} \Big[\ \textit{coeff. of } x^{\,p\,i'-i} \textit{ in } f(x)^{N(k)}\ \Big]\ x^{i'-1}\,\frac{dx}{y^{k_{\mathrm{out}}}}.$$

In other words, the matrix of C in the basis $(\omega_{i,k})$ has, at row (k_{out}, i') and column (k,i), the coefficient of $x^{pi'-i}$ in the trinomial expansion of $f(x)^{N(k)} = (x^d + \sigma x^r + 1)^{N(k)}$ — extracted, as in the classical case $n = 2$, by the multinomial formula

$$[\,x^m\,]\,f(x)^N = \sum_{\substack{\alpha+\beta+\gamma=N \\ d\alpha+r\beta=m}} \binom{N}{\alpha,\beta,\gamma}\,\sigma^\beta.$$

Proof. By Lemma 2.1, $\omega_{i,k} = x^{i-1}\, f(x)^{N(k)} \cdot \big(y^{-k_{\mathrm{out}}}\big)^p\, dx$. Put $h = y^{-k_{\mathrm{out}}}$ and $\omega' = x^{i-1} f(x)^{N(k)}\, dx$: then $\omega_{i,k} = h^p\, \omega'$, and by axiom (i), $C(\omega_{i,k}) = h \cdot C(\omega')$. Now $\omega' = g(x)\, dx$ with $g(x) = x^{i-1} f(x)^{N(k)} \in \mathbb{F}_p[x]$, so $(\star)$ applies: writing $g(x) = \sum_m c_m x^m$,

$$C(\omega') = \sum_{j \geq 1} c_{pj-1}^{1/p}\, x^{j-1}\, dx.$$

Since $c_m \in \mathbb{F}_p$ satisfies $c_m^{1/p} = c_m$ (Fermat's little theorem: we work over $\mathbb{F}_p$ exactly, not over an extension), we obtain

$$C(\omega_{i,k}) = h \cdot \sum_{j \geq 1} c_{pj-1}^{1/p}\, x^{j-1}\, dx = \sum_{j \geq 1} \big[\, \text{coeff. of } x^{pj-1} \text{ in } x^{i-1} f(x)^{N(k)} \,\big]\, x^{j-1}\, \frac{dx}{y^{k_{\mathrm{out}}}}.$$

The coefficient of x^{pj-1} in $x^{i-1} f(x)^{N(k)}$ is that of x^{pj-i} in $f(x)^{N(k)}$. Putting $i' = j$, and observing that $C(\omega_{i,k}) \in H^0(\Omega^1)$ necessarily decomposes on the basis $(x^{i'-1} dx / y^{k_{\mathrm{out}}})_{1 \leq i' \leq g_{k_{\mathrm{out}}}}$ (the Cartier operator preserves holomorphy, a classical fact), the sum restricts to $1 \leq i' \leq g_{k_{\mathrm{out}}}$, which gives the statement. □

Remark 2.3 (Terminology)**.** *Strictly speaking, the matrix M of Theorem 2.2 is the matrix of the* ***Cartier*** *operator, sometimes distinguished in the literature from the* ***Hasse–Witt*** *matrix (which is its adjoint/transpose relative to Serre duality); see Achter–Howe,* Hasse–Witt and Cartier–Manin matrices: a warning and a request*, arXiv:1710.10726, for a warning about this recurring confusion. The distinction has no bearing on any statement below: the rank (raw or stabilised) is invariant under transposition.*

Remark 2.4. *For $n = 2$ the unique block $k = 1$ satisfies $k_{\mathrm{out}} = 1$ (since $1 \cdot p \equiv 1 \pmod 2$ for every odd p) and $N(1) = (p-1)/2$: we recover exactly the classical Cartier–Manin matrix. For $p \equiv 1 \pmod n$, $p^{-1} \equiv 1 \pmod n$ so $k_{\mathrm{out}} = k$ for all k: the matrix is block diagonal (each block closes on itself). For $p \not\equiv 1 \pmod n$ the blocks are permuted along the orbits of $k \mapsto k\, p^{-1} \bmod n$; Theorem 2.2, written with the full $g \times g$ matrix (and not block by block), covers this case without modification.*

3 Theorem B: the p-rank is the stabilised rank, not the raw rank

This is the point we first missed. The construction of Theorem 2.2 has been correct from the start; the error, identified only after a systematic comparison with an independent point count, was to read the p-rank as $\mathrm{rk}(M)$ instead of the stabilised rank under iteration.

Lemma 3.1 (Rank stabilisation – Fitting)**.** *Let $T : V \to V$ be an $\mathbb{F}_p$-linear endomorphism of a finite-dimensional vector space of dimension g. The sequence $(\mathrm{rk}(T^m))_{m \geq 1}$ is non-increasing and stabilises for $m \geq g$: there is $N \leq g$ with $\mathrm{rk}(T^N) = \mathrm{rk}(T^{N+1}) = \cdots$, and for this N,*

$$V = \mathrm{Ker}(T^N) \oplus \mathrm{Im}(T^N),$$

with T bijective on $\mathrm{Im}(T^N)$ and nilpotent on $\mathrm{Ker}(T^N)$.

Proof. $\mathrm{Im}(T^{m+1}) = T(\mathrm{Im}(T^m)) \subseteq \mathrm{Im}(T^m)$ gives the (weak) decrease of the dimensions. A non-increasing sequence of integers in $\{0, \ldots, g\}$ cannot decrease strictly more than g times, so it stabilises at some index $N \leq g$. For such an N, $\mathrm{rk}(T^N) = \mathrm{rk}(T^{2N})$ implies $\mathrm{Im}(T^N) = \mathrm{Im}(T^{2N}) = T^N(\mathrm{Im}(T^N))$, so T^N (hence T) is surjective, hence bijective, on $\mathrm{Im}(T^N)$: this is the classical Fitting lemma. The restriction of T to $\mathrm{Ker}(T^N)$ is nilpotent by definition of $\mathrm{Ker}(T^N)$, and $V = \mathrm{Ker}(T^N) \oplus \mathrm{Im}(T^N)$ follows from the rank–nullity theorem applied to T^N together with $\mathrm{Ker}(T^N) \cap \mathrm{Im}(T^N) = 0$ (if $v \in \mathrm{Im}(T^N) \cap \mathrm{Ker}(T^N)$, write $v = T^N w$; since T^N is bijective on $\mathrm{Im}(T^N)$ and $T^N v = 0$, we get $v = 0$). □

Theorem 3.2 (p-rank = stabilised rank of the Cartier matrix)**.** *Let $M(\sigma)$ be the matrix of the generalised Cartier operator (Theorem 2.2) evaluated at a point $\sigma \in \mathbb{F}_p^\times$. The p-rank of* $\mathrm{Jac}(\mathcal{C}_\sigma)$ *equals the stabilised rank* $\mathrm{rk}(M(\sigma)^N)$ *of Lemma 3.1 (for $N \leq g$ such that the sequence has stabilised),* ***and not*** *necessarily* $\mathrm{rk}(M(\sigma))$*.*

Justification. The p-rank of $\mathrm{Jac}(\mathcal{C})$ is, by definition, the dimension of the étale part of $\mathrm{Jac}(\mathcal{C})[p]$; via the Dieudonné module theory of $\mathrm{Jac}(\mathcal{C})[p]$ (identified with $H^1_{\mathrm{dR}}(\mathcal{C}/\mathbb{F}_p)$ equipped with the Frobenius F and Verschiebung V operators, cf. Oda, *The first de Rham cohomology group and Dieudonné modules*, Ann. Sci. ENS 1969), it is the dimension of the maximal subspace on which F (equivalently, by Serre duality, the Cartier operator $C = V$ on $H^0(\Omega^1)$) acts **bijectively** — that is, exactly $\dim \mathrm{Im}(M^N)$ in the Fitting decomposition of Lemma 3.1, and not $\mathrm{rk}(M)$ in general. The two coincide if and only if M has no non-trivial Jordan block at eigenvalue 0, which is not guaranteed a priori for a block matrix such as that of Theorem 2.2 when $n \geq 3$. □

Remark 3.3. *We do not claim to reprove here Oda's identification of H^1_{dR} with the Dieudonné module: it is a classical theorem that we cite, exactly as the case $n = 2$ relies, without reproving it, on the classical results relating the p-rank to the Cartier–Manin matrix (Manin for the Hasse–Witt matrix; Yui for the product $M\,M^{(p)}\cdots M^{(p^{g-1})}$ over $\mathbb{F}_q$, whose rank is the p-rank; over $\mathbb{F}_p$ this product is M^g). The rank of M alone only gives g minus the a-number. What we establish ourselves, with a complete proof (Lemma 3.1), is that the* "rank" *to take in this classical statement is the stabilised rank under iteration, not the raw rank — a distinction invisible in all the hyperelliptic $n = 2$ cases tested in this project (where stabilisation always occurred at the first iteration, though we cannot assert here that this is always so), but which shows up from $n = 3$ on.*

4 A complete worked example

Take $n = 3$, $d = 4$, $r = 1$, $p = 7$. Then $g = (n-1)(d-1)/2 = 3$, with $g_1 = \lfloor 4/3 \rfloor = 1$ and $g_2 = \lfloor 8/3 \rfloor = 2$. Since $p = 7 \equiv 1 \pmod 3$, the blocks close on themselves ($k_{\mathrm{out}} = k$), with $N(1) = 2$ and $N(2) = 4$.

Case $\sigma = 1$. The block $k = 1$ is 1×1 and zero (no edge in the support). The block $k = 2$ is $M^{(2)} = \left(\begin{smallmatrix} 5 & 5 \\ 4 & 4 \end{smallmatrix}\right) \pmod 7$ (direct computation by the formula of Theorem 2.2, with $f = 1 + x + x^4$ and $N(2) = 4$), of rank 1. The sequence of iterated ranks is $(1, 1, 1, \dots)$: stable from the start. Total stabilised rank $= 0 + 1 = 1$. *Independent check* (point count over $\mathbb{F}_{7^m}$, $m = 1, \dots, 6$, polynomial L reconstructed by the Newton identities): true p-rank = 1. **Exact agreement.**

Case $\sigma = 3$. The block $k = 2$ is (still by the formula of Theorem 2.2) $M^{(2)} = \left(\begin{smallmatrix} 3 & 1 \\ 5 & 4 \end{smallmatrix}\right) \pmod 7$, of rank 1 – **but** $\left(M^{(2)}\right)^2 = 0 \pmod 7$: the sequence of iterated ranks is $(1, 0, 0, \dots)$. Total stabilised rank = 0 (the block $k = 1$ is [0]). *Independent check*: true p-rank = 0. **Exact agreement** – whereas the *non-stabilised* rank $\mathrm{rk}(M) = 1$ would have been wrong.

Remark 4.1. *This second case is precisely the one where a naive reading of the rank (without iteration) produces a wrong result – and it is by confronting this error with an independent point count that Theorem 3.2 was isolated.*

5 Independent verification

The two theorems above were tested, for each configuration (p, n, d, r, σ), against a **second, entirely independent method**: direct point counting of $\mathcal{C}_\sigma$ over $\mathbb{F}_{p^m}$ for $m = 1, \dots, 2g$ (finite fields built via discrete logarithms, without using Cartier theory), reconstruction of the polynomial $L(T) = \sum c_k T^k$ by the Newton identities from the power sums $s_m = p^m + 1 - \#\mathcal{C}(\mathbb{F}_{p^m})$ (exact integer arithmetic, checked to be integral at each step), then reading off the p-rank on the p-adic Newton polygon of $L(T)$ (length of the slope-zero segment).

Regime	Configurations tested	# cases	# agreements
$n = 2$, control	d odd, various (r, σ, p)	3	3
$n = 3$, $p \equiv 1 \pmod 3$	stable blocks, various σ	4	4
$n = 3$, $p \equiv 2 \pmod 3$	exchanged blocks ($k \leftrightarrow 3-k$)	6	6
$n = 4$, $p \equiv 1 \pmod 4$	stable blocks	3	3
$n = 4$, $p \equiv 3 \pmod 4$	cycle of length 2 between blocks	2	2
Total		**18**	**18**

All tested cycles have length ≤ 2; longer cycles are not covered by this check. Reproducible code is provided in the accompanying package (see §20).

6 Catalogue of hidden vanishings and the corank pattern

Definition 6.1 (Hidden vanishing, general case y^n)**.** *For a cycle $(i_0, i_1, \ldots, i_{L-1})$ of the permutation $\sigma(i) = i \cdot p^{-1} \bmod n$, let $\Phi_{i_0}(\sigma) = M^{(i_{L-1})} \cdots M^{(i_1)} M^{(i_0)}$ be the composed return map (square, $g_{i_0} \times g_{i_0}$, where g_{i_0} is the dimension of the starting block). We say that (n, p, d, r) exhibits a **hidden vanishing** at the cycle of i_0 if (a) the bipartite support graph of Φ_{i_0} admits a perfect matching ($\nu(G) = g_{i_0}$) and (b) $\mathrm{rk}\big(\Phi_{i_0}(\sigma)\big) < g_{i_0}$ for **every** $\sigma \in \mathbb{F}_p$ (tested exhaustively, f squarefree).*

Remark 6.2 (Terminology: Fermat-type cases)**.** *In a "Fermat-type" case $\det \Phi \not\equiv 0$ as a polynomial in σ, but $\det \Phi$ vanishes at every $\sigma \in \mathbb{F}_p^\times$ (it is divisible by $\sigma^{p-1} - 1$). These cases satisfy the Definition (b) only because σ is restricted to $\mathbb{F}_p$; they are degenerate exactly at $\sigma^{p-1} = 1$, so they are* not *hidden vanishings in the formal sense ($\det \Phi \equiv 0$), and we keep a distinct name for them. A hand-checkable instance is $(n, p, d, r) = (9, 17, 16, 3)$ and $k = 2$: with $k_1 = 7$, $g_0 = 3$, $g_1 = 12$, $N_0 = 13$, $N_1 = 3$, the 3×3 return map satisfies $\det \Phi \equiv 8(\sigma^{16} - 1) \pmod{17}$ **[V]** (symbolic computation, confirmed numerically for all $\sigma \in \mathbb{F}_{17}^\times$).*

Remark 6.3 (Combinatorial filter)**.** *Condition (a) is checked in negligible time (maximum bipartite matching, Kuhn's algorithm) and **before** any numerical rank computation: by the Leibniz expansion, $\det \Phi_{i_0} = \sum_\pi \mathrm{sgn}(\pi) \prod_i (\Phi_{i_0})_{i,\pi(i)}$ contains only terms indexed by the perfect matchings of the support; if $\nu(G) < g_{i_0}$ no term can be non-zero and $\det \Phi_{i_0} \equiv 0$ **trivially**, with no algebraic coincidence at play. This filter thus eliminates, for free, all non-*hidden *vanishings before the exhaustive sweep of (b), the only costly step — which is what makes a sweep over several thousand families (n, d, r, p) practicable.*

Proposition 6.4 (Survey result)**.** *A combinatorial sweep (filter (a), then exhaustive sweep of (b) on the survivors) covering $n = 3, \ldots, 15$, $d = 2, \ldots, 25$ ($\gcd(d, n) = 1$), all r, primes $p \in \{3, \ldots, 47\}$ ($\gcd(p, n) = 1$, $g \leq 20$), produces **233 confirmed hidden vanishings**. Three of them were cross-checked against an independently written implementation (no shared code), with exact agreement.*

Remark 6.5 (Empirical pattern: corank almost always 1)**.** *Over the 233 cases, the deficit $g_{i_0} - \mathrm{rk}\big(\Phi_{i_0}(\sigma)\big)$ (maximised over σ) equals 1 in 201/233 cases (86%), and this is not an artefact of small blocks: restricting to the cases $g_{i_0} \geq 4$, the deficit is still 1 in 175/206 cases (85%). In other words, when a hidden vanishing occurs it almost always corresponds to a single missing linear relation (corank exactly 1), and not to a deep degeneration — generalising, beyond the isolated case $g_{i_0} = 2$ (where corank 1 is trivial), the outer-product structure observed on the example $(n, d, r, p) = (3, 4, 1, 7)$ (§4). The remaining 14% (deficit 2, and 2 isolated cases of deficit 3 and 4) remain to be explained; an immediate lead is to check whether an iterated outer-product structure (corank k via k independent relations) can be seen directly on the matrices.*

7 Block theory and corank (results proved by hand)

Statuses used throughout: **[D]** proved (hand proof, written in the accompanying note `proofs_blocks.tex`); **[V]** checked numerically on a finite set of cases, not proved; **[O]** open.

Write $k' = k_{\text{out}}$ and, for the block $M^{(k)}$ (of size $g_{k'} \times g_k$), $N = N(k) < p$, $j = p-1-N$, $m = \lfloor dj/p \rfloor$. The entries are $M^{(k)}[b,i] = [x^{pb-i}]f^N$ (Th. 2.2). Since $N < p$, the multinomial coefficients of f^N are non-zero modulo p: the *support* of $M^{(k)}$ is purely combinatorial. Put $L(G) = fG' - jf'G$.

Theorem 7.1 (Kernel of a block, **[D]**)**.** $\operatorname{Ker} M^{(k)}$ *is isomorphic to the space of polynomials* G *of degree* $\le dj$ *such that* $\deg L(G) \le g-1$ *(explicit isomorphism, see* `proofs_blocks.tex`*, Th. 2). There is a basis* (h_M) *of the corresponding subspace (Prop. 6) and*

$$\operatorname{corank} M^{(k)} = m - \operatorname{rk} A, \qquad A_{M,e} = [x^e]\, L(P_M), \quad e \in [g, d-2] \quad (\textit{Th. 7}).$$

Theorem 7.2 (Duality law (K), **[D]**)**.** *The corank* κ_k *of the block* $M^{(k)}$ *(Th. 7.1) satisfies* $\kappa_k = \dim \operatorname{coker} M^{(n-k)}$ *(law (K), Th. 10 of the accompanying note), and*

$$\operatorname{rk} M^{(k)} - \operatorname{rk} M^{(n-k)} = g_k + g_{k'} - (d-1) \quad (\textit{Cor. 11}).$$

In particular, for a self-dual 2-cycle $k_1 = n - k_0$*, the two blocks have the same rank.*

Remark 7.3 (Antecedents)**.** *The block formula of Th. 2.2 already appears in Hase-Liu and Triantafillou (arXiv:1709.02442); see also Sutherland (arXiv:2004.10189), Elkin, Johnston. We do not present it as new. The statements of Th. 7.1–7.2 (description of the kernel by the differential equation* $L(G)$ *and the duality law) were not found in any of the sources consulted, but no zbMATH/MathSciNet-type database could be queried:* ***no claim of novelty****.*

Cycles of length 2: decomposition formula

For a 2-cycle (k_0, k_1) put $M_0 = M^{(k_0)}$, $M_1 = M^{(k_1)}$ and $\Phi = M_1 M_0$ $(g_0 \times g_0)$. Then **[D]**

$$\dim \operatorname{Ker} \Phi = \dim \operatorname{Ker} M_0 + \dim(\operatorname{Im} M_0 \cap \operatorname{Ker} M_1),$$

and the intersection is the radical of a bilinear form $-\operatorname{Res}_0\big(G_1(f\nu' + (j_1+1)f'\nu)\,dx\big)$ on $W_{k_1} \times N_0$ (the two factors live in different spaces: no isotropy test is available).

The case $d = p$

For $d = p$ and a 2-cycle ($N_0 = g_1$, $N_1 = g_0$), put $\beta(j) \equiv -j\, r^{-1} \pmod p$ and $t(j) = (j + r\beta(j))/p$.

Lemma 7.4 (Lemmas A and B, **[D]**)**.** *(A)* $\Phi = \sum_b K_1[a,b]\, \sigma^{\beta(b)} K_0[b,i]\, \sigma^{\beta(i)}$*: each entry of* Φ *factors in two stages, where* K_0, K_1 *are numerical matrices (multinomials) and* σ^β *powers of the parameter. (B) The supports of* K_0, K_1 *are explicit* windows *(bands) in* $(b, \beta(b))$*.*

An index b is called *active* if $\beta(b) \in [1, g_0]$ and the corresponding row of K_0 and column of K_1 are non-zero.

Proposition 7.5 (Active-count criterion, **[D]**)**.** $\operatorname{rk}\Phi \le \#\{\textit{active}\}$*;* $K_1[\cdot, \text{act}]$ *always has full rank, so* Φ *is degenerate* if and only if $K_0[\text{act}, \cdot]$ *is singular. In the* 133/135 *cases studied, the singularity is expressed by a Hall violation visible on a prefix of columns ordered by decreasing* β *(violation sizes:* 1:93*,* 2:32*,* 3:4*,* 4:2*,* 6:1*,* 10:1*).*

For general d, **[D, case by case]**: $\operatorname{rk}\Phi \le \min(\nu(S_0), \nu(S_1), \#\text{active})$, where S_0, S_1 are the combinatorial supports and ν the maximum matching; this criterion explains 39 of the 143 cases of the catalogue.

Cases studied by hand [D]

For the following families (n, p, d, r, k_0, k_1) the degeneracy is proved by hand: $(6, 17, 17, 10, 1, 5)$, $(5, 19, 19, 6, 1, 4)$, $(9, 17, 7, 2, 4, 5)$, $(9, 17, 10, 3, 4, 5)$ (rank $M_0 \le 3$, kernel $(5\sigma^6, 0, 1, 8\sigma^7)$, coker $\simeq M_1$), $(5, 29, 18, 6, 2, 3)$, and $(9, 17, 16, 3, 2, 7)$ with $\det \Phi = 8(\sigma^{16} - 1)$.

8 Classification of the 143 cases of the catalogue [V]

On the 143 cases $(n, p, d, r, \{k_0, k_1\})$ of the survey file (`scan2.json`), exact computation (Sage) gives:

Type	#	Description
Support	39	$\operatorname{rk}\Phi \le \min(\nu(S_0), \nu(S_1), \#\text{act}) < g_0$
Deficient block	49	$\operatorname{rk} M_0 < g_0$ or $\operatorname{rk} M_1 < g_0$ (Th. 7.1, law (K))
Pure matching	44	M_0, M_1 of full rank, Φ deficient: **unexplained**
Fermat	11	$\det\Phi(s)$ divisible by $s^{p-1}-1$, generically invertible
Total	143	

The $39+49+44=132$ cases are "formal" (degeneracy for every σ in an algebraic closure); the 11 Fermat-type cases are degenerate for every squarefree $\sigma \in \mathbb{F}_p^\times$ but not formally.

Fermat mechanism. For $\sigma \in \mathbb{F}_p^\times$ the exponents fold modulo $p-1$; a gauge lemma modulo $p-1$ **[D]** shows that $\det\Phi$ is a polynomial in σ^{p-1} times a power of σ. We observe $\det\Phi(1)=0$ in the 11 cases **[V]**, and a grading modulo $p-1$ in 6/11. Examples: $(9, 17, 16, 3, 2, 7)$, $\det\Phi = 8(\sigma^{16}-1)$; $(8, 5, 21, 8, 3, 7)$, $\det = 2s^9(s^4-1)$. For σ in an extension, the p-rank involves Frobenius-twisted products of Φ: not treated here.

Other observations [V]. (i) 23/143 cases have $\gcd(d, r) > 1$: then $f = h(x^e)$ and Φ decomposes by classes modulo e; counting classes explains only 5 cases (family $d = 18$, Hall type). (ii) Monomial kernels: 19/44 of the kernels of the pure-matching cases, and 46/48 of those of the deficient-block cases. (iii) p-adically, for M_0 integral at $\sigma = 1, 2$, there is an elementary divisor of valuation exactly 1 in 88/90 deficient evaluations. ("$p \mid \det$" alone is tautological and is not a clue.)

Link with the case $n = 2$. In the author's work on y^2, mechanism B is a deployment by monomial gauge $M_{ij} = c_{ij}\sigma^{a_i+b_j}$. In y^n the entries of Φ are not monomial in the 10 hidden cases $d = p$: the structure has two stages (Lemma 7.4).

9 A combinatorial Menger criterion

The general case (any d, any 2-cycle)

For a 2-cycle (k_0, k_1) put $M_0 = M^{(k_0)}$ $(g_1 \times g_0)$, $M_1 = M^{(k_1)}$ $(g_0 \times g_1)$ and $\Phi = M_1 M_0$. Let $\mathcal{G}$ be the oriented graph in three layers $a \to b \to i$, with $a \in [1, g_0]$ (rows of M_1), $b \in [1, g_1]$ (intermediate indices), $i \in [1, g_0]$ (columns of M_0), with an edge $a \to b$ if $M_1[a, b] \neq 0$ and $b \to i$ if $M_0[b, i] \neq 0$. Let Δ be the maximum number of pairwise disjoint paths $a \to b \to i$ (maximum flow with unit vertex capacities).

Theorem 9.1 (Menger, **[D]**). *If $\Delta < g_0$, then $\det\Phi \equiv 0$ (identically in σ), for every d.*

Proof. By Cauchy–Binet, $\det\Phi = \sum_S A_S B_S$, where S runs over the subsets of g_0 indices of $[1, g_1]$, $A_S = \det M_1[\,\cdot\,, S]$ and $B_S = \det M_0[S, \cdot\,]$. By Leibniz, $A_S \neq 0$ requires a perfect matching of the rows a into S in the support of M_1, and $B_S \neq 0$ a perfect matching of S onto the columns i in the support of M_0. If such an S exists, the two matchings compose into g_0 disjoint paths $a \to b \to i$, so $\Delta = g_0$. By contraposition, $\Delta < g_0$ implies $A_S B_S = 0$ for all S, hence $\det\Phi = 0$. For a cycle of length L one replaces the two stages by L stages. □

This criterion is strictly finer than the perfect-matching criterion on the support of Φ (Remark of §8): the Boolean product of the supports may admit a perfect matching although no injective factorisation $a \to b \to i$ exists. On the 91 cases of the sweep ($p \equiv -1 \pmod n$, $d \le 25$, $p \le 47$, blocks of full rank, support of Φ with a perfect matching), Theorem 9.1 explains 37 **of the 67 formal cases** (status **[V]** for the counts, **[D]** for the implication); in the 37 cases the flow deficiency is exactly 1.

The case $d = p$: closed criterion

Proposition 9.2 ([**D**]). *Let $d = p$ and (k_0, k_1) a 2-cycle. Then $N(k_1) = g_0$ and $N(k_0) = g_1$.*

Proof. $p^2 \equiv 1 \pmod n$ gives $k_1 \equiv k_0 p \pmod n$ with $0 < k_1 < n$; the Euclidean division $k_0 p = n\, g_0 + k_1$ (since $g_0 = \lfloor k_0 p/n \rfloor$) gives $g_0 = (k_0 p - k_1)/n = N(k_1)$. Likewise for g_1. □

Put $c(v) = \lceil rv/p \rceil$, $w(v) = v - c(v)$, $\beta(v) = p\, c(v) - rv$.

Lemma 9.3 (supports by intervals, [**D**]). *For $d = p$: $M_1[a, b] \neq 0$ if and only if there is $v \in [1, g_0]$ with $b = \beta(v) \leq g_1$ and $a \in I_v := [c(v),\ g_0 - w(v)]$; $M_0[b, i] \neq 0$ if and only if there is $v \in [1, g_1]$ with $i = \beta(v) \leq g_0$ and $b \in J_v := [c(v),\ g_1 - w(v)]$.*

Proof. $M_1[a, b] = [x^{pa-b}] f^{g_0}$ is a sum of multinomials $\binom{g_0}{u, v, \cdot} \sigma^v$ over $pu + rv = pa - b$, $u + v \leq g_0$; they are non-zero modulo p because $g_0 < p$. Since $1 \leq b \leq g_1 < p$, we have $b \equiv -rv \pmod p$; $v = 0$ is excluded (then $b \equiv 0$), and $v \leq g_0 < p$, so $b = \beta(v)$ for a unique v, with $pa - b = pu + rv$, i.e. $u = a - c(v) \geq 0$ and $u + v \leq g_0$, that is $a \leq g_0 - w(v)$. Same argument for M_0 (with $N(k_0) = g_1$). □

Since c and w are non-decreasing, the intervals I_v (resp. J_v) are **nested decreasing**: $I_{v+1} \subseteq I_v$.

Theorem 9.4 (Menger criterion for $d = p$, [**D**]). *Let $V_I = \{v(i) = (-i\, r^{-1}) \bmod p : 1 \leq i \leq g_0\}$, $v_m = \min V_I$ and $B^* = \{\beta(v) : 1 \leq v \leq g_0,\ \beta(v) \leq g_1,\ \beta(v) \in J_{v_m}\}$. Then $\Delta = g_0$ if and only if:*

1. *every $v \in V_I$ satisfies $v \leq g_1$;*
2. *$|B^*| = g_0$;*
3. *for every $v_0 \in V_I$: $\#\{v \in V_I : v \geq v_0\} \leq \#(J_{v_0} \cap B^*)$.*

Proof. Stage 1 ($a \to b$). For a set V'' of columns of M_1 of smallest index v_0, $N(V'') = I_{v_0}$ (nesting), of length $g_0 - v_0 + 1$; the distinct v lie in $[v_0, g_0]$, so $|V''| \leq g_0 - v_0 + 1$: Hall's condition holds, all usable columns get matched, and the matching saturates the rows a if and only if $|B^*| = g_0$ (condition 2). *Stage 2 ($i \to b$).* Same argument with the $J_v \cap B^*$: Hall's condition for the g_0 columns i reduces to $\#\{v \in V_I : v \geq v_0\} \leq \#(J_{v_0} \cap B^*)$ for every $v_0 \in V_I$ (condition 3); condition 1 is the non-vacuity of the columns. *Composition.* If both stages saturate, we have two bijections $a \leftrightarrow B^* \leftrightarrow i$ (since $|B^*| = g_0$), whose composition gives g_0 disjoint paths. Conversely g_0 disjoint paths give the two matchings, hence the conditions. □

Check [V]. On 6104 cases with $d = p$ and a 2-cycle ($n \leq 12$, $p \leq 47$, all r, all k_0), the formula of Lemma 9.3 reproduces exactly the supports of M_0, M_1 (6104/6104), and the criterion of Theorem 9.4 coincides with the exact flow (6104/6104). Among these cases: 5710 of support type ($\nu(\Phi) < g_0$), 107 of hidden Menger type, 287 with $\Delta = g_0$ (formal, Fermat or zero-minor candidates); all 107 fail at the second stage (85 at both stages, 22 at the second only) and none has three layers.

Remark 9.5 (Beyond $d = p$). *For $d \neq p$ the support of a column is a union of intervals (a point (u, v) of the lattice $du + rv = pa - b$, two parameters): over* 1500 *blocks, the columns are contiguous in* 68/68 *cases $d = p$, but only* 353/811 *(M1) and* 226/811 *(M0) for $d < p$, and* 366/622 *and* 84/622 *for $d > p$. There is thus no interval criterion: for $d \neq p$ Theorem 9.1 remains an exact and fast test (flow), without a closed formula. The* 7 *three-layer cases (an obstruction that appears neither at stage 1 alone nor at stage 2 alone) are all cases $d > p$.*

10 The observed mechanisms

On the 91 pure-matching cases of the sweep $p \equiv -1 \pmod n$, $d \le 25$, $p \le 47$ (blocks of full rank, $\nu(\Phi) = g_0$): the exact computation of $\det \Phi(s)$ as a polynomial in s gives 67 *formal* cases ($\det \Phi \equiv 0$) and 24 Fermat-type cases ($\det \Phi \neq 0$, zero on $\mathbb{F}_p^\times$). The corank equals 1 in all 91 cases *of this family* $p \equiv -1 \pmod n$ (outside this family it is not true: §13). The 67 formal cases are distributed as follows ([**V**]):

Mechanism	Cases	Status
1. Menger obstruction ($\Delta < g_0$)	37	[**D**] implication, [**V**] count
2. Zero minor with non-empty support	16 tested, 12 not tested	arithmetic coincidence
3. Cancellation between non-zero terms	2	by-hand example: §11

Mechanism 1. See §9; on the 37 cases the flow deficiency is 1; 30 are explained by a Hall condition at a single stage (13 at both, 9 stage 2 only, 8 stage 1 only) and 7 require three layers. For 7 cases whose cut contains only vertices b, the columns used by M_1 are essentially $b_a = a(p-r)$ (term $(\sigma x^r)^a$, or $a(p-d)$ for the term $(x^d)^a$); the last b_a fall outside the window of rows of M_0, and Hall fails (example $(6, 17, 17, 10, 1, 5)$: M1 only reaches $b = 7$ for $a = 1$ and $b \in \{7, 14\}$ for $a = 2$, M0 has no row 14).

Mechanism 2. When $N(k_1)$ is small, M_1 is almost monomial and forces S; B_S is then a small determinant of multinomials. Reading through the *cube* of multinomials $C[u, v, w] = N!/(u!v!w!)$ ($u + v + w = N$): four points A, B, C, D with $A + D = B + C$ (automatic) give a vanishing 2×2 minor if and only if $R = \prod_c B_c!C_c!/(A_c!D_c!) \equiv 1 \pmod p$. Example $(8, 23, 23, 17, 1, 7)$: $A = (3, 4, 13)$, $B = (0, 8, 12)$, $C = (9, 4, 7)$, $D = (6, 8, 6)$, $R = 588/13$, $588 - 13 = 25 \cdot 23$. Example $(7, 13, 22, 14, 1, 6)$: $A = (7, 1, 3)$, $B = (5, 4, 2)$, $C = (10, 0, 1)$, $D = (8, 3, 0)$, $R = 20/7$, $20 - 7 = 13$. In both cases $R - 1$ equals exactly p times a unit. Statistically ([**V**], monomial entries) the proportional pairs of rows of M_0 are 239/7166 for $g_0 = 2$ (expected $1/p$: 299) and the pairs forced by M_1 are 18/740 (expected 27); for $g_0 = 3$, 1722 zero determinants for $t = 1$ distinct exponent (expected 5100), 39 for $t = 2$ (66), 0 for $t \ge 3$. There is no excess of cancellations compared with chance: *no arithmetic law is apparent* for this mechanism ([**O**]).

Common kernel of layers. The classical idea: if $\Phi(s) = \sum_t C_t s^t$ and $W = \bigcap_t \ker C_t \neq 0$, then $\operatorname{rk} \Phi(s) \le g_0 - \dim W$ for every s ([**D**], trivial). On our 67 formal cases, this criterion (raw coefficients or after the smallest monomial gauge) detects only 11, of which 10 are already detected by Menger; it adds a single case. The converse is false ($\left(\begin{smallmatrix} 1 & t \\ t & t^2 \end{smallmatrix}\right)$).

Fermat. 24 cases, $\det \Phi \neq 0$ as a polynomial, zero on $\mathbb{F}_p^\times$ (§12 for the structure and a by-hand example, $(8, 3, 5, 1, 5)$); examples $\det = 8(\sigma^{16} - 1)$ for $(9, 17, 16, 3, 2, 7)$. None has a common kernel, none is of Menger type (flow $= g_0$): this is an effect of the folding of exponents modulo $p - 1$, not yet explained in detail ([**O**]).

11 A degree law and two complete by-hand proofs

This section contains only [**D**] statements: each computation is done by hand, without a machine. Notation: for a 2-cycle (k_0, k_1), $M_0 = M^{(k_0)}$ $(g_1 \times g_0)$, $M_1 = M^{(k_1)}$ $(g_0 \times g_1)$, $\Phi = M_1 M_0$, with $M^{(k)}[b, i] = [x^{pb-i}]\, f^{N(k)}$ and $N(k) < p$. For $S \subset [1, g_1]$ of cardinality g_0, A_S (resp. B_S) is the minor of M_1 on the columns S (resp. of M_0 on the rows S); by Cauchy–Binet, $\det \Phi = \sum_S A_S B_S$. We write $T = g_0(g_0 + 1)/2$ and $\Sigma S = \sum_{b \in S} b$.

Proposition 11.1 (degree law, **[D]**). *Let S be as above. Expand $A_S B_S$ by Leibniz and each entry into its multinomial terms $c\,x^{du+rv}\sigma^v$ (with $du + rv = pb - i$ for the entry (b, i)). For a term of the product, let $U = \sum u$ and $V = \sum v$ be the totals (V is the degree in σ). Then*

$$d\,U + r\,V = (p-1)\,(T + \Sigma S).$$

Proof. A term of A_S chooses a bijection $\pi : [1, g_0] \to S$ and an entry $(a, \pi(a))$ of M_1 for each a; a term of B_S chooses a bijection $\rho : S \to [1, g_0]$ and an entry $(b, \rho(b))$ of M_0 for each $b \in S$. The sum of the exponents of x is

$$\sum_{a=1}^{g_0} \big(pa - \pi(a)\big) + \sum_{b \in S} \big(pb - \rho(b)\big) = pT - \Sigma S + p\,\Sigma S - T = (p-1)(T + \Sigma S),$$

since π and ρ are bijections ($\sum_a \pi(a) = \sum_{b \in S} b = \Sigma S$ and $\sum_b \rho(b) = T$). But this sum also equals $dU + rV$. □

Corollary 11.2 (**[D]**). *Two terms of the same degree V in σ, coming from S and S', satisfy $d\,(U - U') = (p-1)(\Sigma S - \Sigma S')$; in particular $d \mid (p-1)(\Sigma S - \Sigma S')$. For a single S, $rV \equiv (p-1)(T + \Sigma S) \pmod{d}$: ΣS determines V modulo $d/\gcd(d, r)$.*

Example A: a cancellation between two non-zero terms

Take $(n, p, d, r, k_0) = (8, 3, 7, 2, 5)$: the curve $y^8 = x^7 + \sigma x^2 + 1$ over $\mathbb{F}_3$, of genus $g = (7)(6)/2 = 21$. Then $k_1 = 5 \cdot 3^{-1} = 5 \cdot 3 \equiv 7 \pmod 8$ (and $7 \cdot 3 \equiv 5$, so it is indeed a 2-cycle), $g_0 = \lfloor 35/8 \rfloor = 4$, $g_1 = \lfloor 49/8 \rfloor = 6$, $N(k_0) = (7 \cdot 3 - 5)/8 = 2$, $N(k_1) = (5 \cdot 3 - 7)/8 = 1$.

Blocks. $f = 1 + \sigma x^2 + x^7$; $M_1[a, b] = [x^{3a-b}]f$ gives

$$M_1 = \begin{pmatrix} \sigma & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Next $f^2 = 1 + 2\sigma x^2 + \sigma^2 x^4 + 2x^7 + 2\sigma x^9 + x^{14}$ (all coefficients are $\not\equiv 0 \pmod 3$), and $M_0[b, i] = [x^{3b-i}]f^2$ gives

$$M_0 = \begin{pmatrix} 2\sigma & 0 & 1 & 0 \\ 0 & \sigma^2 & 0 & 2\sigma \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2\sigma & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Which S matter. The rows of M_1 have supports $\{1,3\}, \{4,6\}, \{2\}, \{5\}$: $A_S \neq 0$ requires $2, 5 \in S$, one element of $\{1,3\}$ and one of $\{4,6\}$, i.e. $S \in \{\{1,2,4,5\}, \{1,2,5,6\}, \{2,3,4,5\}, \{2,3,5,6\}\}$. The rows of M_0 have supports $\{1,3\}, \{2,4\}, \{2\}, \{3\}, \{1\}, \{4\}$. For $S = \{1,2,4,5\}$: row 5 forces column 1, row 4 column 3, and row 1 has no column left: $B_S = 0$. For $S = \{2,3,5,6\}$: rows $3, 5, 6$ force $2, 1, 4$ and row 2 has no column left: $B_S = 0$. Two S remain.

$S = \{1,2,5,6\}$. For A_S: row $1 \mapsto$ column 1 (σ), row $2 \mapsto 6$, row $3 \mapsto 2$, row $4 \mapsto 5$, i.e. the permutation of positions $(1, 4, 2, 3)$, a 3-cycle, of sign $+$: $A_S = \sigma$. For B_S (rows $1, 2, 5, 6$): $1 \mapsto 3$, $2 \mapsto 2$ (σ^2), $5 \mapsto 1$, $6 \mapsto 4$, permutation $(3, 2, 1, 4)$, a transposition, of sign $-$: $B_S = -\sigma^2$. Hence $A_S B_S = -\sigma^3$.

$S = \{2,3,4,5\}$. A_S: rows $1 \mapsto 3$, $2 \mapsto 4$ (σ), $3 \mapsto 2$, $4 \mapsto 5$, positions $(2, 3, 1, 4)$, 3-cycle: $A_S = \sigma$. B_S (rows $2, 3, 4, 5$): $2 \mapsto 4$ (2σ), $3 \mapsto 2$ (2), $4 \mapsto 3$ (2σ), $5 \mapsto 1$ (1), permutation $(4, 2, 3, 1)$, a transposition: $B_S = -8\sigma^2$. Hence $A_S B_S = -8\sigma^3$.

Proposition 11.3 ([D]). *For* $(n,p,d,r,k_0)=(8,3,7,2,5)$*:* $\det\Phi=-\sigma^3-8\sigma^3=-9\sigma^3$ *(identity in* $\mathbb{Z}[\sigma]$*), hence* $\det\Phi\equiv 0 \pmod 3$ *for every* $\sigma\in\mathbb{F}_3$*. Moreover* M_0, M_1 *have full rank* 4 *for* $\sigma\neq 0$ *(rows* $3,4,5,6$ *of* M_0 *have supports* $\{2\},\{3\},\{1\},\{4\}$*; the rows of* M_1 *have supports* $\{1,3\},\{4,6\},\{2\},\{5\}$*), and two* S *are admissible, so that* $\Delta=g_0=4$*: the obstruction is not of Menger type. For* $\sigma=2$ *(the only smooth value in* $\mathbb{F}_3^\times$*: for* $\sigma=1$*,* $x=1$ *is a double root of* f*),* $\operatorname{rk}\Phi<4$*. Furthermore the degree law holds: for both* S*,* $\Sigma S=14$*,* $T=10$*,* $(p-1)(T+\Sigma S)=48=7\cdot 6+2\cdot 3$ *with* $(U,V)=(6,3)$*.*

(That $f=1+2x^2+x^7$ is squarefree in $\mathbb{F}_3[x]$: $f'=x(1+x^5)$; $f(0)=1$; if $x^5=-1$ then $x^7=-x^2$ and $f=1+x^2$, hence $x^2=-1$, $x^4=1$, $x^5=x$, so $x=-1$, $x^2=1\neq -1$: contradiction.) This is a case of *hidden vanishing*: the support of Φ has a perfect matching, the blocks have full rank, and yet $\operatorname{rk}\Phi<g_0$. The mechanism is a *cancellation* $-1-8=-9\equiv 0$ between two non-zero products of the same degree. Note that $-9\equiv 0$ even modulo 9: it is a coincidence of small p, not yet a law.

Example B: a Fermat-type case, $\det\Phi\neq 0$ but zero on $\mathbb{F}_p^\times$

Take $(n,p,d,r,k_0)=(8,3,5,1,5)$: $y^8=x^5+\sigma x+1$ over $\mathbb{F}_3$, genus 14. Here $k_1=7$, $g_0=\lfloor 25/8\rfloor=3$, $g_1=\lfloor 35/8\rfloor=4$, $N(k_0)=2$, $N(k_1)=1$, $f=1+\sigma x+x^5$. $M_1[a,b]=[x^{3a-b}]f$:

$$M_1=\begin{pmatrix}0&\sigma&1&0\\1&0&0&0\\0&0&0&1\end{pmatrix}.$$

$f^2=1+2\sigma x+\sigma^2x^2+2x^5+2\sigma x^6+x^{10}$ and $M_0[b,i]=[x^{3b-i}]f^2$:

$$M_0=\begin{pmatrix}\sigma^2&2\sigma&1\\2&0&0\\0&0&2\sigma\\0&1&0\end{pmatrix}.$$

Then $\Phi=M_1M_0$ has rows $\sigma\cdot(2,0,0)+(0,0,2\sigma)=(2\sigma,0,2\sigma)$, $(\sigma^2,2\sigma,1)$ and $(0,1,0)$. Expanding along the last row,

$$\det\Phi=-\det\begin{pmatrix}2\sigma&2\sigma\\\sigma^2&1\end{pmatrix}=-(2\sigma-2\sigma^3)=2\sigma(\sigma^2-1).$$

Proposition 11.4 ([D]). *For* $(8,3,5,1,5)$*:* $\det\Phi=2\sigma(\sigma^2-1)\in\mathbb{F}_3[\sigma]$*, a* non-zero *polynomial, which vanishes for every* $\sigma\in\mathbb{F}_3^\times=\{1,2\}$*. For* $\sigma=2$ *(the only smooth value:* $f=1+2x+x^5$*,* $f'=2(1+x^4)$*, and* $x^4=-1$ *with* $f(x)=0$ *would give* $1+x=0$*, i.e.* $x^4=1\neq -1$*),* $\Phi=\begin{pmatrix}1&0&1\\1&1&1\\0&1&0\end{pmatrix}$ *satisfies row 2 − row 1 = row 3:* $\operatorname{rk}\Phi=2<3$*, corank* 1*. The blocks have rank* 3 *and the support of* Φ *has a perfect matching.*

Reading. In example A the determinant is identically zero; in example B it is not, but it is divisible by $\sigma^{p-1}-1=\sigma^2-1$. These are the two types of the classification of §10 (formal / Fermat), each with a representative that can be checked by hand. In both cases $p=3$: this is the value of p most favourable to coincidences ($1+2\equiv 0$), and we deduce no law from it.

12 Structure of the Fermat-type cases and new sweeps [V]

Coverage correction. The sweeps of this section and of §13 run over *all* $r\in[1,d-1]$. An intermediate version of our scripts only examined $r\leq d-r$, assuming the symmetry $x\mapsto 1/x$;

this symmetry is **false** for y^n with $\gcd(d,n) = 1$ (the change of variable turns the family into $y^n = t^e(\dots)$ with $e \equiv -d \pmod n$, a cover also ramified at $t = 0$): for instance $(8,3,7,2,5)$ is hidden whereas $(8,3,7,5,5)$ is not (15 different labels on 101 randomly drawn pairs $(r, d-r)$). The earlier catalogues (91, 143, 6104 cases) ran over all r.

Complementary sweeps (all r). (i) $n \le 10$, $p \le 31$, $d \le 25$: 10415 cases (not deduplicated by k_0/n), *no anomaly* (the check "flow $< g_0 \Rightarrow \det\Phi \equiv 0$" is a check of the code, the theorem being proved); 54 Menger cases, 24 formal cases outside Menger, 15 of Fermat type, 186 undecided ($g_0 > 8$). (ii) $p \in \{3,5,7\}$, $n \le 24$, $d \le 40$ (8200 distinct cases): 50 Menger, 7 formal outside Menger, 30 of Fermat type, 353 undecided ($g_0 > 8$). The union of the two, deduplicated by $(p, d, r, k_0/n)$, contains 21 formal cases outside Menger, which the Cauchy–Binet computation separates as follows:

Mechanism	Distinct cases	Examples
Zero minor (each $A_S B_S = 0$)	14	$(8,23,23,17,1)$, $(5,29,16,13,2)$, $(8,5,9,1,3)$
Cancellation between non-zero products	7	$(8,3,7,2,5)$, $(8,3,11,4,5)$, $(3,11,20,6,1)$

Four of the seven cancellations have $p = 3$, one $p = 5$, two $p = 11$: they are concentrated on small p, consistent with a coincidence of the type $1 + 2 \equiv 0$, though we have no law. (The domains of these tallies differ from that of the census of §16; they are not reconciled.)

The Fermat-type cases. Among 39 distinct Fermat-type cases with $g_0 \le 8$ (deduplicated), 38 have a $\det\Phi$ divisible by $s^{p-1} - 1$; the exception $(8,5,9,6,3)$ has $\det\Phi = 3s^2 + 4s^5 + 2s^8 + s^{11}$, which vanishes at $\sigma = 1, 2, 4$ (values where f is squarefree) but not at $\sigma = 3$ (where f has a square factor). So *the correct statement* is "$\det\Phi$ vanishes at all smooth σ", and not "is divisible by $s^{p-1} - 1$" (which holds in 38/39 cases). The multiplicity of $s^{p-1} - 1$ is 1 in 36 cases, 2 in two cases, $(12,5,13,1,7)$ and $(24,5,11,7,19)$, and 0 in the exception above; see §13.

Binomial cases. Seven cases have a $\det\Phi$ with two monomials, $c\, s^a(s^{p-1} - 1)$ (in the sense that the two monomials have exponents $p-1$ apart and coefficients opposite modulo p). In the first six of the table, each monomial comes from a single S (after two internal cancellations for the last), and the two S differ *by a single exchanged index*:

Case (n,p,d,r,k_0)	g_0	exchange $b \to b'$	exponents V	coefficients mod p
$(8,3,5,1,5)$	3	$2 \to 3$	1, 3	1, 2
$(8,3,13,5,5)$	8	$1 \to 6$	10, 12	1, 2
$(24,5,11,7,13)$	5	$2 \to 6$	7, 11	3, 2
$(8,5,21,8,3)$	7	$4 \to 17$	9, 13	3, 2
$(12,7,17,2,5)$	7	$5 \to 7$	13, 19	6, 1
$(8,13,23,7,3)$	8	$2 \to 9$	21, 33	7, 6
$(9,17,16,3,2)$	3	(a single S gives both exponents)	0, 16	9, 8

That the gap between exponents is exactly $p-1$ is nothing deep: a binomial that vanishes on $\mathbb{F}_p^\times$ requires $e_1 \equiv e_2 \pmod{p-1}$ and the exponents remain small. The two facts to explain are (a) why only two terms remain and (b) why the coefficients are opposite; we have no answer ([**O**]).

What cannot be asserted. Over the *integers* (unreduced multinomials), the two coefficients are exactly opposite for $(8,3,5,1,5)$ $(-2,+2)$ and $(8,3,13,5,5)$ $(+16,-16)$, but not in the others: ratio -6 for $(8,5,21,8,3)$ $(\equiv -1 \pmod 5)$, and integer ratio $7134627 \equiv -1 \pmod{17}$ for $(9,17,16,3,2)$. We therefore claim *no involution* between the two S: only the case $p = 3$ with two admissible S is clean.

13 Outside the family $p \equiv -1 \pmod n$: the corank

§10 records corank 1 in the 91 cases of the family $p \equiv -1 \pmod n$. For a 2-cycle it suffices that $p^2 \equiv 1 \pmod n$; the complementary family $p \not\equiv \pm 1 \pmod n$ behaves differently.

Theorem 13.1 (**[D]** multiplicity and corank)**.** *Let $\Phi(s) \in M_g(\mathbb{F}[s])$ ($\mathbb{F}$ a field) and $s_0 \in \mathbb{F}$ with $\dim \operatorname{Ker} \Phi(s_0) = c$. Then $(s - s_0)^c$ divides $\det \Phi(s)$.*

Proof. Let $P \in GL_g(\mathbb{F})$ whose last c columns span $\operatorname{Ker} \Phi(s_0)$. The last c columns of $\Phi(s)P$ are $\big(\Phi(s) - \Phi(s_0)\big)$ applied to these vectors; their entries are divisible by $s - s_0$. By multilinearity of the determinant in these c columns, $(s - s_0)^c$ divides $\det(\Phi P) = \det \Phi \cdot \det P$, and $\det P \in \mathbb{F}^\times$. □

Consequence: if $\det \Phi \not\equiv 0$ vanishes at σ_0 with a corank c there, its multiplicity there is $\geq c$. This is what is observed: in $(12, 5, 13, 1, 7)$, $g_0 = 7$, $g_1 = 11$, we find (**[V]**) $\det \Phi = 2\sigma^7(\sigma^4 - 1)^2$ in $\mathbb{F}_5[\sigma]$ and a corank 2 for every $\sigma \in \mathbb{F}_5^\times$; the neighbouring case $(12, 5, 13, 6, 7)$ has corank 1.

Survey [V]. For $p^2 \equiv 1 \pmod n$, $p \not\equiv \pm 1$, with $p \leq 31$, $n \leq 40$, $d \leq 30$, $g_0 \leq 12$, all r, deduplicated by $(p, d, r, k_0/n)$: 16331 cases of deficient rank for every smooth σ, among which 449 *pure* (blocks of full rank and support of Φ with a perfect matching, hence hidden vanishings). The corank of these 449 cases is: 1 for 388, 2 for 52, 3 for 8, 5 for 1. The computation is identical in Python and under Sage (on the restricted range $r \leq d - r$ that we had first used, 8782 cases of which 187 pure, the two environments give the same counts). The cross table by g_0 and $N_0 + N_1$ versus p:

	corank 1	corank ≥ 2	share of corank ≥ 2
$g_0 \leq 5$, $N_0 + N_1 < p$	75	2	2.6 %
$g_0 \leq 5$, $N_0 + N_1 \geq p$	62	4	6.1 %
$g_0 \geq 6$, $N_0 + N_1 < p$	79	2	2.5 %
$g_0 \geq 6$, $N_0 + N_1 \geq p$	172	53	23.6 %

Corank ≥ 2 is thus concentrated where $g_0 \geq 6$ *and* $N_0 + N_1 \geq p$. Neither condition alone suffices, and we have neither a law nor an explanation (**[O]**). Consequence for the statement of §10: "corank 1" is an observed property of the family $p \equiv -1 \pmod n$, not of all 2-cycles.

14 Generic support: Conjecture G [V]/[O]

Let $E \subset \{0, \ldots, d\}$ with $0, d \in E$ and $f_c = \sum_{e \in E} c_e x^e$ with indeterminate coefficients c_e. Since $N(k) < p$, the multinomials do not vanish modulo p: the support of $M^{(k)}$ is determined by the sums of $N(k)$ terms of E (in the sense of sumsets), and the flow $\Delta(E)$ of §9 follows.

Proposition 14.1 (**[D]**)**.** *If $\Delta(E) < g_0$ then $\det \Phi_c \equiv 0$ in $\mathbb{F}_p[c]$.*

(Same proof as Theorem 9.1: only the support matters.)

*Conjecture G **[O]**.* For $|E| \geq 4$: $\det \Phi_c \not\equiv 0$ if and only if $\Delta(E) = g_0$.

Evidence. The direction "⇒" is the proposition. For "⇐": ≈ 1230 trials with random coefficients on supports E of size ≥ 4 (including 474 "adversarial" supports: arithmetic progressions, gcd > 1), no counterexample. For $|E| = 3$ the converse is *false*: all the "zero minor" and "cancellation" cases above are trinomials with $\Delta = g_0$ and $\det \Phi \equiv 0$. An attempted proof by grading (Proposition 11.1) and tropical weights (Monge arrays) runs into the fact that rectangles contained in one segment of the hull give the same total monomial; binomial minors can then vanish modulo p, exactly as in mechanism 2. **The conjecture is not proved.**

Adding a fourth monomial [V]. Replacing f by $f+\tau x^s$ (τ generic) lifts the Menger obstruction for 34 of the 37 Menger cases of the family $p \equiv -1 \pmod n$, for every s; the three exceptions occur when s is a small multiple of r modulo d. The rank is then full in 57 cases out of 60 draws; the 3 draws that fall short are not explained.

15 Remarks: y^2 and the link with the complex route

The case $n = 2$. The Menger graph reduces to a Hall condition on the support of Φ, which is the very definition of hidden vanishing: Theorem 9.1 is then empty. An earlier sweep ($p \le 59$, d odd ≤ 29) found 8 hidden cases for y^2, none with $d = p$; they come under the monomial gauge known for y^2 ([**V**]).

Hypergeometric dictionary [O]. The periods $\int x^i\,dx/y^k$ of the family are expressed by hypergeometric series in $z = \sigma^m$, $m = d/\gcd(d, r)$, which split according to the class of v mod m; the Hasse–Witt matrix is a truncation of such series (A-hypergeometric series, Adolphson–Sperber, arXiv:1601.05127). Their Theorem 2.3 (generic invertibility of the Hasse–Witt determinant under a support condition, by a convex-hull argument) has the same form as Conjecture G. We have read only excerpts: it is not used in our proofs, and we do not know whether it implies Conjecture G.

16 Version 4.1: second iterate, kernel of a block, exact forms and rank cuts

This section adds proved results ([**D**]), numerical checks ([**V**]) and open points ([**O**]). It also corrects a point of scope: unless stated otherwise, the statements "for every σ" concern $\sigma \in \mathbb{F}_p$ (see Lemma 16.1). The arguments about exact forms are standard (the kernel of the Cartier operator is the space of exact forms); we claim no novelty on these points.

The second iterate

On $\mathbb{F}_p[x]\,dx$ the Cartier operator is $C\big(\sum c_m x^m dx\big) = \sum_j c_{pj-1}\,x^{j-1}dx$. It is $\mathbb{F}_p$-linear and satisfies $C(g^p\omega) = g\,C(\omega)$ and $C(dg) = 0$. With the notation of Th. A, $M_0[b, i] = [x^{pb-i}]f^{N_0}$ is the matrix of $h \mapsto C(f^{N_0}h\,dx)$ on polynomials $h = \sum_i h_i x^{i-1}$ of degree $< g_0$ towards those of degree $< g_1$; likewise for M_1 and N_1.

Lemma 16.1 ([**D**], second iterate)**.** *Let $\sigma \in \mathbb{F}_p$, $f = 1 + \sigma x^r + x^d$ and $M = pN_1 + N_0$. Then $\Phi = M_1M_0$ is the matrix of $h \mapsto C^2(f^M h\,dx)$, i.e. $\Phi[a, i] = [x^{p^2a-i}]f^M$.*

Proof. $C(f^{N_1}\,C(f^{N_0}h\,dx)) = C\big(C(f^{pN_1}f^{N_0}h\,dx)\big)$ because $f^{N_1}C(\omega) = C\big((f^{N_1})^p\omega\big)$ (property $C(g^p\omega) = gC(\omega)$ with $g = f^{N_1}$, with coefficients in $\mathbb{F}_p$). Then one reads the coefficient of x^{p^2a-1} in $f^M x^{i-1}$. □

Remark 16.2 (**Frobenius twist**)**.** *For $\sigma \in \mathbb{F}_q \setminus \mathbb{F}_p$ the Cartier operator is semilinear ($C(c\omega) = c^{1/p}C(\omega)$) and the correct product involves a factor twisted by Frobenius (this is the twisted product of Achter–Howe). The statements "$\det\Phi \equiv 0$ for every σ" of this document are therefore to be read for $\sigma \in \mathbb{F}_p$. Menger's theorem and the statements about vanishing minors, which depend only on the support, are not concerned. Section 17 gives a numerical test of what happens outside $\mathbb{F}_p$.*

The kernel of a block is the image of a first-order differential operator

For $f = 1 + \sigma x^r + x^d$ and $N \ge 1$, put $L_N q = f\,q' + (N+1)f'\,q$; we have $L_N q = (qf^{N+1})'/f^N$.

Theorem 16.3 ([**D**])**.** *Let f be squarefree in $\mathbb{F}_p[x]$ and $N < p$. A polynomial h satisfies $C(f^N h\,dx) = 0$ if and only if there is $q \in \mathbb{F}_p[x]$ with $h = L_N q$. The kernel of the block M (window $\deg h < g$) is therefore $\{L_N q : \deg L_N q < g\}$.*

Proof. ($\Leftarrow$) $f^N h = (q f^{N+1})'$ is exact, so $C(f^N h\,dx) = 0$.

($\Rightarrow$) A polynomial form is killed by C if and only if it has no monomial x^{pk-1}, i.e. if and only if it is exact: $f^N h = g'$ with $g \in \mathbb{F}_p[x]$ (defined modulo $\mathbb{F}_p[x^p]$). At each root ρ of f (a simple root), write $g = \sum_m c_m (x-\rho)^m$; since $f^N \mid g'$, we have $m\,c_m = 0$ for $1 \le m \le N$, and m is invertible in $\mathbb{F}_p$ because $N < p$. Hence $g \equiv g(\rho) \pmod{(x-\rho)^{N+1}}$.

Let $A = \mathbb{F}_p[x]/(f)$, an étale algebra; the Frobenius $a \mapsto a^p$ of A is bijective. Let $w_0 \in \mathbb{F}_p[t]$ be such that the class of $w_0(\bar{x})$ is $\mathrm{Fr}^{-1}(\bar{g})$, and put $w = w_0$. Then $w(x^p) = w_0(x)^p \equiv g \pmod f$, so $w(\rho^p) = g(\rho)$ for every root. Now $x^p - \rho^p = (x-\rho)^p$, so $w(x^p) \equiv w(\rho^p) = g(\rho) \pmod{(x-\rho)^p}$, in particular modulo $(x-\rho)^{N+1}$ (since $N+1 \le p$). Thus $g - w(x^p)$ is divisible by $(x-\rho)^{N+1}$ for each root, hence by f^{N+1} (f monic with distinct roots): $g = w(x^p) + q f^{N+1}$. As $w(x^p)' = 0$, we get $f^N h = g' = f^N (f q' + (N+1) f' q)$, i.e. $h = L_N q$. □

[V] On 223 smooth cases where M_0 is deficient ($p \le 13$, $d \le 15$, $g_0 \le 8$), the dimension of the kernel coincides with that of $\{L_N q : \deg L_N q < g_0\}$; it suffices to take $\deg q < d(p-1-N)$ (this is the bound of the companion note: L_N coincides with $L(G) = fG' - j f' G$, $j = p-1-N$, whose window is $\deg G \le dj$, for windows $g \le d$); a smaller bound would wrongly give a smaller kernel (we first made this truncation error). **[V]** The bound was tested on 2000 random cases with 0 mismatches (`verify_kernel_theorem.py`).

A recurrence and the twin columns

Lemma 16.4 (**[D]**, recurrence)**.** *Let $f^N = \sum a_e x^e$. For every $e \in \mathbb{Z}$: $e\,a_e + \sigma\,(e-(N+1)r)\,a_{e-r} + (e-(N+1)d)\,a_{e-d} = 0$.*

Proof. Compare the coefficients of x^{e-1} in $f\,(f^N)' = N\,f' f^N$: on the left $e\,a_e + \sigma(e-r)a_{e-r} + (e-d)a_{e-d}$, on the right $N\sigma r\,a_{e-r} + N d\,a_{e-d}$. □

Write $c_m[b] = a_{pb-m}$ (column m, $m \in \mathbb{Z}$). With $e = pb - i$ we have $e \equiv -i \pmod p$, whence a relation *with constant coefficients* between columns:

$$-i\,c_i - \sigma\,(i+(N+1)r)\,c_{i+r} - (i+(N+1)d)\,c_{i+d} \equiv 0 \pmod p.$$

Corollary 16.5 (**[D]**, twin columns)**.** *If $i \equiv -(N+1)d \pmod p$, $1 \le i$, $i + r \le g_0$ and $i \not\equiv 0 \pmod p$, then $c_i = -\sigma\,\dfrac{i+(N+1)r}{i}\,c_{i+r}$ in M_0 modulo p. Hence $\mathrm{rk}\,M_0 < g_0$ and $\det \Phi \equiv 0 \pmod p$ for every σ.*

Examples. (a) $(d, r, p, k) = (4, 1, 7, 2)$, $n = 3$ (cycle of length 1, $N = 4$, $g = 2$): $f = 1 + \sigma x + x^4$, f^4 gives $M = \begin{pmatrix} 12\sigma^2 & 12\sigma \\ 4\sigma & 4 \end{pmatrix}$ (coefficients $12 = 4!/(1!\,2!\,1!)$ at x^6, 12 at x^5, $4 = 4!/(0!\,1!\,3!)$ at x^{13}, $4 = 4!/(1!\,0!\,3!)$ at x^{12}), column $1 = \sigma\cdot$ column 2 and $\det M = 48\sigma^2 - 48\sigma^2 = 0$ exactly in $\mathbb{Z}[\sigma]$; the corollary predicts it: $i \equiv -5 \cdot 4 = -20 \equiv 1 \pmod 7$. (b) $(n, p, d, r, k_0) = (8, 19, 11, 3, 5)$: $N = 16$, $i \equiv -17 \cdot 11 \equiv 3$, partner $i + r = 6$, and $-(3 + 17\cdot 3)/3 \equiv 1$, so $c_3 = \sigma c_6$: the equality is even exact in $\mathbb{Z}[\sigma]$ (the terms $(u, v, w) = (m, m+1, 15-2m)$ of column 3 and $(m+1, m, 15-2m)$ of column 6 have the same multinomial coefficient). The corollary thus explains five of the seven single-twin-column cases of M_0 in the census below: $(8, 11, 13, 4, 5)$, $(8, 11, 17, 1, 5)$, $(12, 17, 19, 4, 7)$, $(8, 19, 11, 3, 5)$, $(12, 31, 17, 3, 5)$. The two other twins $(8, 11, 19, 14, 5)$ (columns 3, 8, shift $d - r$) and $(8, 29, 15, 1, 3)$ (columns 1, 3, ratio σ^2) come from other relations of the same family or from a composition; we have not proved them.

The module of exact forms: observed laws

L_N is $\mathbb{F}_p[t]$-linear for $t = x^p$, so $\mathrm{Im}\,L_N$ is a submodule of $\mathbb{F}_p[x] = \bigoplus_{s=0}^{p-1} \mathbb{F}_p[t] x^s$. Let Λ_N be the set of leading degrees of its non-zero elements; the dimension of the kernel of the block of window g is $\#\{\ell \in \Lambda_N : \ell < g\}$ (Th. 16.3).

- **[D]** No element of $\operatorname{Im} L_N$ has leading degree $\equiv -1 - Nd \pmod p$: the leading term of $f^N h = g'$ has degree $\deg h + Nd$, and a derivative has no monomial x^{pk-1}.
- **[V]** Over 108 736 situations ($p \le 17$, $d \le 21$, all r, all $N < p$, σ such that f is squarefree): the module has rank $p-1$; if $\nu_s = s + pk_s$ is the smallest leading degree of class s, then $\sum_s k_s = \lfloor Nd/p \rfloor$ (conservation law, not proved); the profile (ν_s) is independent of σ in 9731 of the 10 302 combinations (p, d, r, N) having at least two smooth σ (94.5 %) and depends on σ in 571.
- **[V]** For the binomial ($\sigma = 0$, $c = (N+1)d \not\equiv 0$): the profile has as missing leading degrees the class $-1 - Nd$ and the starting vertices $v_0 < d - 1$ such that $(-1 - v_0)d^{-1} \bmod p < (-1 - c - v_0)d^{-1} \bmod p$ (checked on 1307 cases out of 1307). Example $(p, d, N) = (7, 4, 4)$: class 4 and starts $1, 2$.
- **[V]** For the trinomial, the simple rule "first cancellation of a coefficient" (first leading degree $= a_0 + r - 1$ or $a_0 - 1$ with $a_0 = -(N+1)d \bmod p$) holds in only 20 % of cases: no closed formula ([**O**]).

The rank cut

Proposition 16.6 ([**D**], Menger with true ranks)**.** *For every partition* $\{1, \dots, g_1\} = S \sqcup S^c$*:* $\operatorname{rk} \Phi \le \operatorname{rk} M_1[:, S] + \operatorname{rk} M_0[S^c, :]$*. More generally, for every invertible diagonal matrix* D *of size* g_1*,* $\operatorname{rk}(M_1 D M_0) \le \operatorname{rk} M_1[:, S] + \operatorname{rk} M_0[S^c, :]$*.*

Proof. $M_1 D M_0 = M_1[:, S] D_S M_0[S, :] + M_1[:, S^c] D_{S^c} M_0[S^c, :]$; the first term has rank $\le \operatorname{rk} M_1[:, S]$, the second rank $\le \operatorname{rk} M_0[S^c, :]$, and rank is subadditive. □

Menger's theorem is the same bound with the rank replaced by the maximum matching of the support. This inequality is elementary and probably classical; we claim no novelty. The version with D is used in §17.

Census [V]. Domain: $p \le 31$, $3 \le n \le 12$, $p^2 \equiv 1 \pmod n$, $p \not\equiv 1 \pmod n$, $d \le 25$, $\gcd(d, n) = 1$, all r, $2 \le g_0 \le 12$, pairs (k_0, k_1) with $k_1 > k_0$. We keep the cases where $\det \Phi \equiv 0$ *identically in* $\mathbb{Z}[\sigma]$ (test at a random σ modulo $2^{61} - 1$), with a perfect matching in the support of Φ: 33 cases, i.e. 27 distinct cases once the cases $(n, k_0) \sim (n', k_0')$ with the same k_0/n are identified (our earlier sweeps did not make this identification). Three groups are concerned: $(3, 17, 19, 6, 1), (6, 17, 19, 6, 2), (9, 17, 19, 6, 3), (12, 17, 19, 6, 4)$; $(3, 29, 17, 5, 1), (6, 29, 17, 5, 2), (12, 29, 17, 5, 4)$; $(5, 19, 23, 8, 1), (10, 19, 23, 8, 2)$.

Class	distinct cases	mechanism
Twin columns of M_0 ($c_i = \sigma^k c_j$, $k \in \{\pm 1, \pm 2\}$)	7	Cor. 16.5 for 5 of them
Twin rows of M_1	3	$(3, 11, 20, 13, 1)$, $(9, 17, 20, 6, 2)$, $(8,$
Three-column relation of M_0	1	$(8, 19, 13, 7, 5)$: $-16\sigma c_1 + c_7 + 18\sigma$
Mixed: $\operatorname{rk} M_0 = \operatorname{rk} M_1 = g_0$ but a cut S with bound $< g_0$	13	Prop. 16.6, bound $= \operatorname{rk} \Phi$ in all cas
Cancellations between non-zero terms	3	$(8, 3, 11, 4, 5)$, $(8, 3, 11, 7, 5)$, $(8, 5, 1$

In the three twin-row cases, the script `census_exact_zero.py` shows that $\operatorname{rk} M_0 = g_0 - 1$ as well; the table records the mechanism we identified for M_1, and we have not identified one for the deficiency of M_0 there. The script `census_exact_zero.py` reproduces this census (27 distinct cases: 11 with a deficient block, 13 with a rank cut, 3 without a cut).

For the 13 mixed cases, the cut was found by enumerating all S ($g_1 \le 14$) or by local search; it is checked at one numerical $\sigma \in \mathbb{F}_p$ per case, except for $(12, 23, 13, 10, 5)$ treated below for every σ. This census concerns cancellation *exact over* $\mathbb{Z}$; it is not reconciled with the table "14 zero minors / 7 cancellations" of §12, obtained on a smaller domain with another criterion. Cases that vanish only modulo p are not in this census.

A complete by-hand proof: the mixed case $(n,p,d,r,k_0)=(12,23,13,10,5)$

Here $k_1=5\cdot 23^{-1}\equiv 7 \pmod{12}$, $g_0=\lfloor 65/12\rfloor=5$, $g_1=\lfloor 91/12\rfloor=7$, $N_0=(7\cdot 23-5)/12=13$, $N_1=(5\cdot 23-7)/12=9$, $f=1+\sigma x^{10}+x^{13}$ over $\mathbb{F}_{23}$. The entries are $\sum \frac{N!}{u!v!w!}\sigma^v$ over $u+v+w=N$, $10v+13w=e$, with $e=23b-i$ for M_0 ($N=13$) and $e=23a-b$ for M_1 ($N=9$).

Step 1: rows $b=1,2,3$ of M_0 are supported on columns $\{3,4\}$. Row $b=1$: exponents $18,\dots,22$, of which only $20=10\cdot 2$ has the form $10v+13w$ ($v+w\le 13$); entry $\frac{13!}{11!\,2!}\sigma^2=78\sigma^2$ in column 3. Row 2: exponents $41,\dots,45$, only $43=3\cdot 10+13$ fits: $\frac{13!}{9!\,3!\,1!}\sigma^3=2860\sigma^3$ in column 3. Row 3: exponents $64,\dots,68$; only $66=4\cdot 10+2\cdot 13$ (column 3, $25740\sigma^4$) and $65=5\cdot 13$ (column 4, 1287) fit. Hence $\operatorname{rk} M_0[\{1,2,3\},:]\le 2$.

Step 2: $\operatorname{rk} M_1[:,\{4,5,6,7\}]\le 2$ *modulo* 23. Column 5 of M_1 is zero (exponents $18,41,64,87,110$: none of the form $10v+13w$ with $v+w\le 9$). Columns $4,6,7$ have as non-zero entries (rows $a=1,\dots,5$):

$$c_4=(0,0,126,252\sigma,36\sigma^2),\quad c_6=(0,126\sigma^4,504\sigma^5,252\sigma^6,0),\quad c_7=(0,84,630\sigma,756\sigma^2,84\sigma^3),$$

for example $126=\frac{9!}{4!\,5!}$ (exponent $65=5\cdot 13$), $252=\frac{9!}{2!\,1!\,6!}$ (exponent $88=10+6\cdot 13$), $36=\frac{9!}{0!\,2!\,7!}$ ($111=20+7\cdot 13$), etc. One checks, row by row,

$$13\sigma^5\,c_4+7\,c_6+\sigma^4\,c_7\equiv 0 \pmod{23},$$

since the integer sums are: row 2: $7\cdot 126+84=966=23\cdot 42$; row 3: $13\cdot 126+7\cdot 504+630=5796=23\cdot 252$; row 4: $13\cdot 252+7\cdot 252+756=5796=23\cdot 252$; row 5: $13\cdot 36+84=552=23\cdot 24$; row 1: all entries are zero. Hence $\operatorname{rk} M_1[:,\{4,5,6,7\}]\le 2$.

Step 3: conclusion. With $S=\{4,5,6,7\}$ and $S^c=\{1,2,3\}$, Prop. 16.6 gives $\operatorname{rk}\Phi\le 2+2=4<5=g_0$, hence $\det\Phi\equiv 0 \pmod{23}$ for every $\sigma\in\mathbb{F}_{23}$. (The two bounds are those of single factors, so by the diagonal version of Prop. 16.6 the same conclusion holds for every product M_1DM_0 with D invertible diagonal; see §17.)

The support of Φ nevertheless admits a perfect matching (so this is not a Menger case): for $\sigma\neq 0$, the following paths $a\to b\to i$ use only entries that are non-zero modulo 23 and are disjoint: $1\to 3\to 4$, $5\to 1\to 3$, $2\to 7\to 5$, $3\to 6\to 1$, $4\to 4\to 2$ (entries $M_1[1,3]=13\sigma^2$, $M_0[3,4]=22$; $M_1[5,1]=9\sigma$, $M_0[1,3]=9\sigma^2$; $M_1[2,7]=15$, $M_0[7,5]=13$; $M_1[3,6]=21\sigma^5$, $M_0[6,1]=12\sigma^2$; $M_1[4,4]=22\sigma$, $M_0[4,2]=2\sigma^9$). This is a hidden vanishing, with $\operatorname{rk}\Phi=4$ checked numerically.

17 The variant $y^n=x^d+\sigma x^r+t$ and parameters outside $\mathbb{F}_p$

Let $t\in\mathbb{F}_p^\times$ and $f_t=t+\sigma x^r+x^d$. The blocks $M_0(\sigma,t)$, $M_1(\sigma,t)$ are defined by the same formulas, with $t^u\sigma^v$ in place of σ^v in the multinomial sum, and $\Phi(\sigma,t)=M_1M_0$ is the honest Cartier product of the $\mathbb{F}_p$-curve $y^n=f_t$ (no twist is needed, since all coefficients lie in $\mathbb{F}_p$).

Proposition 17.1 ([**D**], scaling)**.** *Let $\lambda\in\overline{\mathbb{F}}_p$ with $\lambda^d=t$ and $\sigma'=\sigma\lambda^{r-d}$. Then $M_j(\sigma,t)[b,i]=t^{N_j}\lambda^{-pb}\cdot\lambda^i\,M_j(\sigma',1)[b,i]$ (for $j=0$; for $j=1$ replace b,i by a,b): each block is a row scaling times a column scaling of the block at $(\sigma',1)$. Consequently $\Phi(\sigma,t)=D_1\,M_1(\sigma',1)\,D\,M_0(\sigma',1)\,D_2$ with D_1,D_2 invertible diagonal and $D=\operatorname{diag}(\lambda^{(1-p)b})_b$. In particular, if $\lambda\in\mathbb{F}_p$ (e.g. if t is a d-th power in $\mathbb{F}_p^\times$), then $D=1$ and $\operatorname{rk}\Phi(\sigma,t)=\operatorname{rk}\Phi(\sigma',1)$ with $\sigma'\in\mathbb{F}_p$.*

Proof. $f_t(\lambda x)=t+\sigma\lambda^r x^r+tx^d=t\,f_{\sigma'}(x)$, hence $[x^e]f_t^N=\lambda^{-e}t^N[x^e]f_{\sigma'}^N$; put $e=pb-i$. The product is formed with λ^b (from M_1) times λ^{-pb} (from M_0), which gives the inner factor $\lambda^{(1-p)b}$. □

For $\lambda\notin\mathbb{F}_p$ (i.e. t not a d-th power in $\mathbb{F}_p$), σ' in general lies outside $\mathbb{F}_p$ and M_1DM_0 is the $\mathbb{F}_p$-rational model of the Frobenius-twisted product at σ' (we do not re-derive here the exact convention of the twist). So the honest $\mathbb{F}_p$-computation of $\Phi(\sigma,t)$ for t not a d-th power gives the rank at a parameter $\sigma'\notin\mathbb{F}_p$, without having to implement $\mathbb{F}_q$ arithmetic.

Corollary 17.2 ([**D**], persistence of factor-wise obstructions)**.** *If the deficiency* $\operatorname{rk}\Phi < g_0$ *is proved by (i) a Menger obstruction* ($\Delta < g_0$)*, or (ii) a rank deficiency of* M_0 *or of* M_1 *proved by a column relation valid for* σ *in any extension (such as Cor. 16.5), or (iii) a cut (Prop. 16.6) whose two bounds are bounds on the ranks of the single factors, then the same deficiency holds for* $M_1 D M_0$ *for every invertible diagonal* D*, and hence for* $\Phi(\sigma,t)$*, for all* (σ,t) *and also at parameters* σ' *outside* $\mathbb{F}_p$*. (For (ii) and (iii) the relations among columns hold for* σ *in any extension; for Cor. 16.5 this is because Lemma 16.4 does not use* $\sigma \in \mathbb{F}_p$*.)*

This applies to the twin-column, twin-row and mixed-cut classes of the census; it does not apply to cancellations between non-zero terms, whose mechanism is a numerical coincidence between several products $A_S B_S$ and may not survive a change of the inner diagonal. **Warning (correction).** It is *not* true that $\det\Phi(\sigma,1) \equiv 0$ in $\mathbb{Z}[\sigma]$ implies $\det\Phi(\sigma,t) \equiv 0$ in $\mathbb{Z}[\sigma,t]$ by a weight argument: the total t-exponent of a term of $A_S B_S$ is affine in the σ-degree V but with an offset that depends on S (through ΣS, Prop. 11.1), so setting $t = 1$ can merge terms of different S that stay separate for $t \neq 1$.

Sweep [V]. Script `tvariant.py` (in the package). Domain: $p \le 31$, $3 \le n \le 12$, $p^2 \equiv 1 \pmod n$, $p \not\equiv 1 \pmod n$, $3 \le d \le 25$, $\gcd(d,n) = 1$, $\gcd(d, p-1) > 1$, all r, $2 \le g_0 \le 12$, $k_1 > k_0$. We keep the cases (not deduplicated by k_0/n) where, at $t = 1$, the rank of Φ is $< g_0$ for every smooth $\sigma \in \mathbb{F}_p^\times$ and the union of the supports has a perfect matching: 59 cases. For each we compute $\operatorname{rk}\Phi(\sigma,t)$ for all smooth $(\sigma,t) \in (\mathbb{F}_p^\times)^2$ and split by t a d-th power (then $\sigma' \in \mathbb{F}_p$) or not.

	cases	t a d-th power: deficient for all	t not a d-th power: deficient for all
$\det\Phi \equiv 0$ over $\mathbb{Z}$	32	32	32
vanishing only modulo p	27	27	13 (14 lose the deficiency at some (σ,t))

(Exactness over $\mathbb{Z}$ is tested at a random σ modulo $2^{61}-1$.) So: (1) for all 32 cases with an identically vanishing determinant over $\mathbb{Z}$, the deficiency persists at every tested (σ,t), including parameters outside $\mathbb{F}_p$; this is consistent with Cor. 17.2, which explains persistence for the factor-wise mechanisms, but we have not checked that each of these 32 cases is of such a type. (2) The vanishings that occur only modulo p are $\mathbb{F}_p$-specific in more than half the cases: 14 of 27 lose the deficiency at some (σ,t) with t not a d-th power. Sanity check: for t a d-th power the deficiency never disappears, as Prop. 17.1 predicts; the identity $\operatorname{rk}\Phi(\sigma,t) = \operatorname{rk}\Phi(\sigma',1)$ was also checked on 274 random draws (0 discrepancy). For the 9 cases tabulated in §16 one has $\gcd(d,p-1) = 1$, so every t is a d-th power and the variant adds nothing there. [**O**]: a proof or a mechanism for the 14 cases that lose the deficiency; whether every exact cancellation over $\mathbb{Z}$ persists outside $\mathbb{F}_p$.

Open after version 4.2

- The 3 arithmetic cancellations ($p = 3, 5$), with no rank cut: [**O**].
- Closed formula for the profile of $\operatorname{Im} L_N$ for the trinomial, proof of the law $\sum k_s = \lfloor Nd/p \rfloor$ and of the independence in σ: [**O**].
- Independence in σ of the rank cut (checked at one numerical σ per case, for every σ only for $(12, 23, 13, 10, 5)$ and, by Cor. 17.2, for the factor-wise mechanisms): [**O**].
- Cycles of length ≥ 3; exact convention of the twisted product off $\mathbb{F}_p$; novelty with respect to the literature: not checked.

18 Bibliography consulted

- Hase-Liu, Triantafillou, arXiv:1709.02442; Sutherland, arXiv:2004.10189: Cartier–Manin matrix of superelliptic curves by blocks (direct antecedent of Th. 2.2).

- Elkin, *The rank of the Cartier operator on cyclic covers of the projective line*, arXiv:0708.0431: decomposition of $H^0(\Omega^1)$ into eigenspaces D_i, permutation σ of the blocks, rank bounds $\sum \min(2\lfloor d_i/p \rfloor, d_{\sigma(i)}) \leq \mathrm{rk} \leq \sum \min(d_i, d_{\sigma(i)})$. Close antecedent of the block structure and of our per-block rank bounds; we read it only through its abstract and excerpts.

- Yi (student of W. Li), talk *a-number and p-rank of superelliptic curves*, Washington University: the p-rank of the curves $y^l = f(x)$ as the stabilised rank of the Cartier–Manin matrix; results not read.

- Blache, *Hasse–Witt matrices for polynomials, and applications* (Rend. Sem. Mat. Univ. Padova 145, 2021): Hasse–Witt matrix depending on the set of exponents; full text not read. Blache, arXiv:2607.21033 (2026): generic ordinarity of abelian covers (the abstract states that it generalises results of Bouw, Li–Mantovan–Pries–Tang, Moonen and Dollarhide; we read only the abstract and identify no gap in it).

- Blanco-Chacón, Boix, Fordham, Yilmaz, arXiv:1712.05763 (level and ordinarity); Varchenko, arXiv:1806.03289; Pries, arXiv:1806.04654 and talks; Li–Mantovan–Pries–Tang and Tang (special families of cyclic covers); Achter–Howe, arXiv:1710.10726 (the Cartier–Manin product over $\mathbb{F}_q$ is a product *twisted* by Frobenius); Ibukiyama–Katsura–Oort (1986).

- Adolphson, Sperber, arXiv:1601.05127: Hasse–Witt matrix as a truncated A-hypergeometric series, generic invertibility under a support condition (Th. 2.3); the statement assumes a full support (a hypothesis that our families do not satisfy in general); excerpts only, not used in our proofs.

- Moonen, and Bouw et al., arXiv:1805.06914 (Ex. 4.5): signature of an orbit $\{\tau, \tau^*\}$ and expected stabilised rank $\min(g_0, g_1)$; read partially, not used in our proofs.

- Earlier work of the author on $y^2 = x^d + \sigma x^r + 1$ (case $n = 2$).

None of the sources consulted (abstracts or excerpts) treats the Menger criterion, nor the *hidden* vanishings (support with a perfect matching but rank $< g$) for y^n; this does not prove their absence from the literature.

19 What this theory does not claim

- $\gcd(d, n) = 1$ **is a hypothesis**; $\gcd(d, n) > 1$ is not treated.

- **No general theory**: Th. 9.4 is closed for $d = p$ and 2-cycles only; for $d \neq p$ Th. 9.1 gives a test, not a formula. Cycles of length ≥ 3, polynomials f not squarefree, $\sigma \in \mathbb{F}_{p^s}$ (see §17 for a numerical test only): not treated ([**O**]).

- **Coverage correction**: the symmetry $r \leftrightarrow d - r$ is false; the sweeps of §12–13 run over all r, but the Cauchy–Binet check is limited to $g_0 \leq 8$ (light sweeps) and to $\binom{g_1}{g_0} \leq 20000$; 186 cases (first sweep, not deduplicated) and 353 cases (second sweep) have $g_0 > 8$ and are not decided.

- **Conjecture G** (§14): not proved; the cross table of the corank (§13) is an observation, not a law.

- **Open**: the Fermat mechanism (24 cases); the 7 three-layer cases ($d > p$); the 12 untested cases of mechanism 2; the 2 cases of cancellation between terms; the arithmetic coincidences of mechanism 2; two numerical coincidences $(8, 23, 23, 17, 1, 7)$, $(12, 19, 12, 5, 11)$ of the catalogue of 143.

- **Coverage**: the numbers in this file come from distinct sweeps (catalogue of 233 cases of §8, 143 classified cases, 91 cases of the sweep $p \equiv -1 \pmod{n}$, 6104 cases $d = p$); their exact reconciliation has not been done. Point counting only checks $g \leq 4$.
- **Retractions of version 4.1 and 4.2**: the statistics "76/68 cases where $\ker M_0 > \operatorname{Im} L_N$" and "310/262" are withdrawn (truncation artefact on the degree of q, and missing "square-free" filter in older tests modulo p); the 33 exact cases contained duplicates (27 distinct); the conservation law $\sum k_s = \lfloor Nd/p \rfloor$ and the independence of the profile in σ are not proved; the 3 arithmetic cancellations and the three-column relation of $(8, 19, 13, 7, 5)$ remain without mechanism; the older tallies (14 zero minors, 7 cancellations) and the census $p \leq 31$, $g_0 \leq 12$ have different domains; the claim that $\det \Phi(\sigma, 1) \equiv 0$ over $\mathbb{Z}$ implies the same for $\det \Phi(\sigma, t)$ by a weight argument is withdrawn (§17).
- **Novelty not claimed**: no zbMATH/MathSciNet-type database has been consulted, and several cited texts were read only through their abstract. The block formula and the per-block rank bounds have antecedents (Hase-Liu, Sutherland, Elkin).
- Th. 3.2 relies on Oda (cited, not reproved).
- No Galois connection and no closed formula of the type of Part III for y^n.

20 Availability

The accompanying package (archive `vanissements_yn_scripts.zip`, with a `README.md` and a `LICENSE`) contains, all in English:

- `verify_hand_proofs.py`: the by-hand examples A and B and the proof of $(12, 23, 13, 10, 5)$ (§11, §16);
- `verify_kernel_theorem.py`: Theorem 16.3 (random cases, with the truncation control);
- `verify_recurrence_twins.py`: Lemma 16.4, Corollary 16.5 and examples (a), (b);
- `verify_exact_forms_laws.py`: the profile of $\operatorname{Im} L_N$ ([D] and [V] laws);
- `cut_search.py` and `census_exact_zero.py`: Prop. 16.6 and the census of §16 (27 distinct cases);
- `tvariant.py`: Prop. 17.1 and the sweep of §17 (it also serves as the common library of the other scripts);
- `verify_yn_pure_python.py`: independent point-count check of Theorems A and B (18 configurations);
- `vanissements_yn.sage` (all-in-one test bench: blocks, stabilised rank, Menger flow, exact determinant, Cauchy–Binet, common kernel of layers, fourth monomial, criterion $d = p$, sweep; also valid Python) and `corank_survey.sage` (corank survey of §13);
- `proofs_blocks.tex`: the note of complete proofs cited in §7.

The large sweeps quoted in §8–13 (233 and 143 cases, 6104 cases, 16331 cases) were run with earlier versions of the scripts and are not all re-runnable one-by-one from the package; the package reproduces the new results of versions 4.1 and 4.2 and the by-hand proofs.